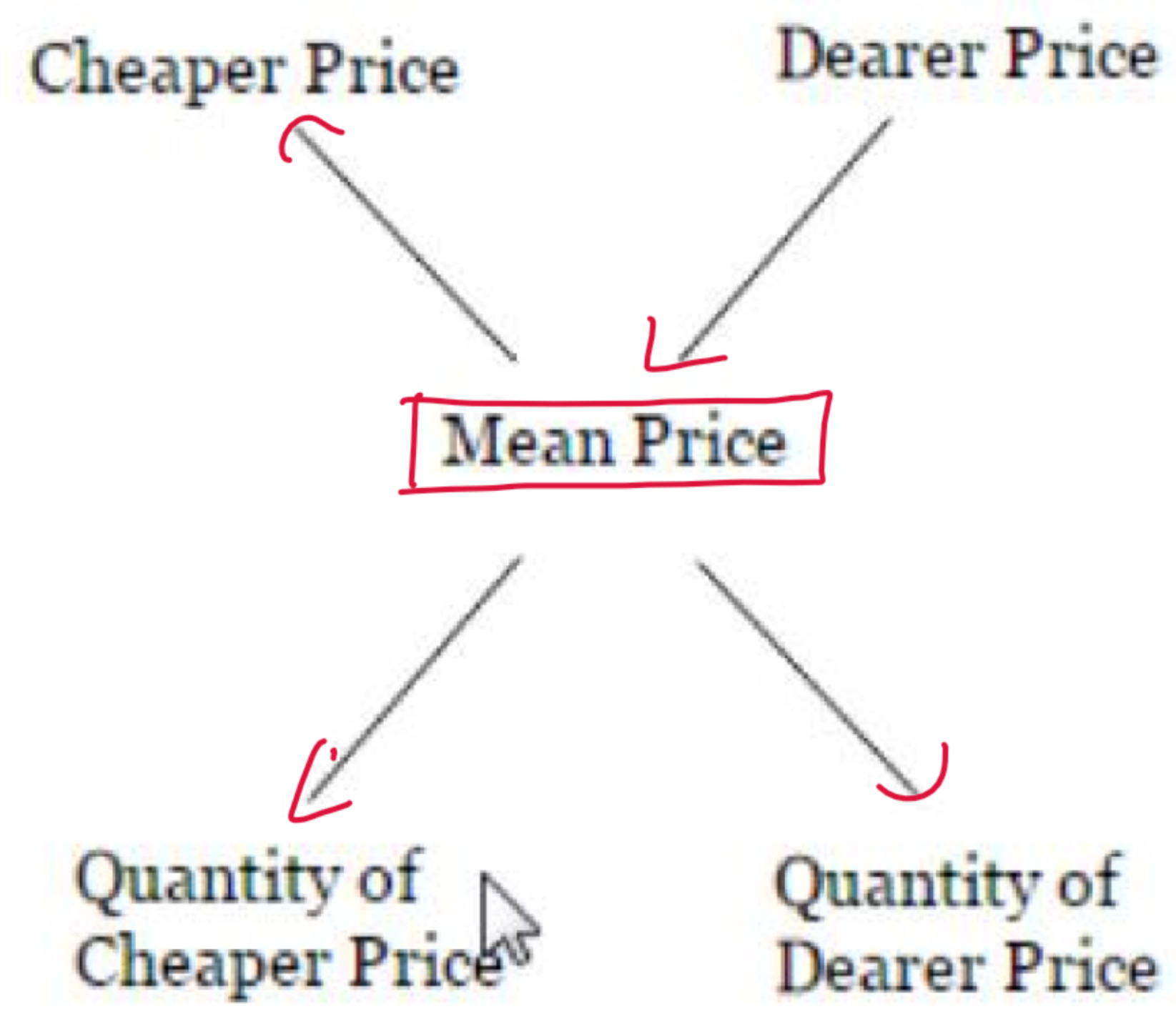


## Mixture and Allegation

Rule of Allegation  $[D > m > C]$   $D = \text{Dearer}$ ,  $m = \text{Mean}$ ,  $C = \text{cheaper}$



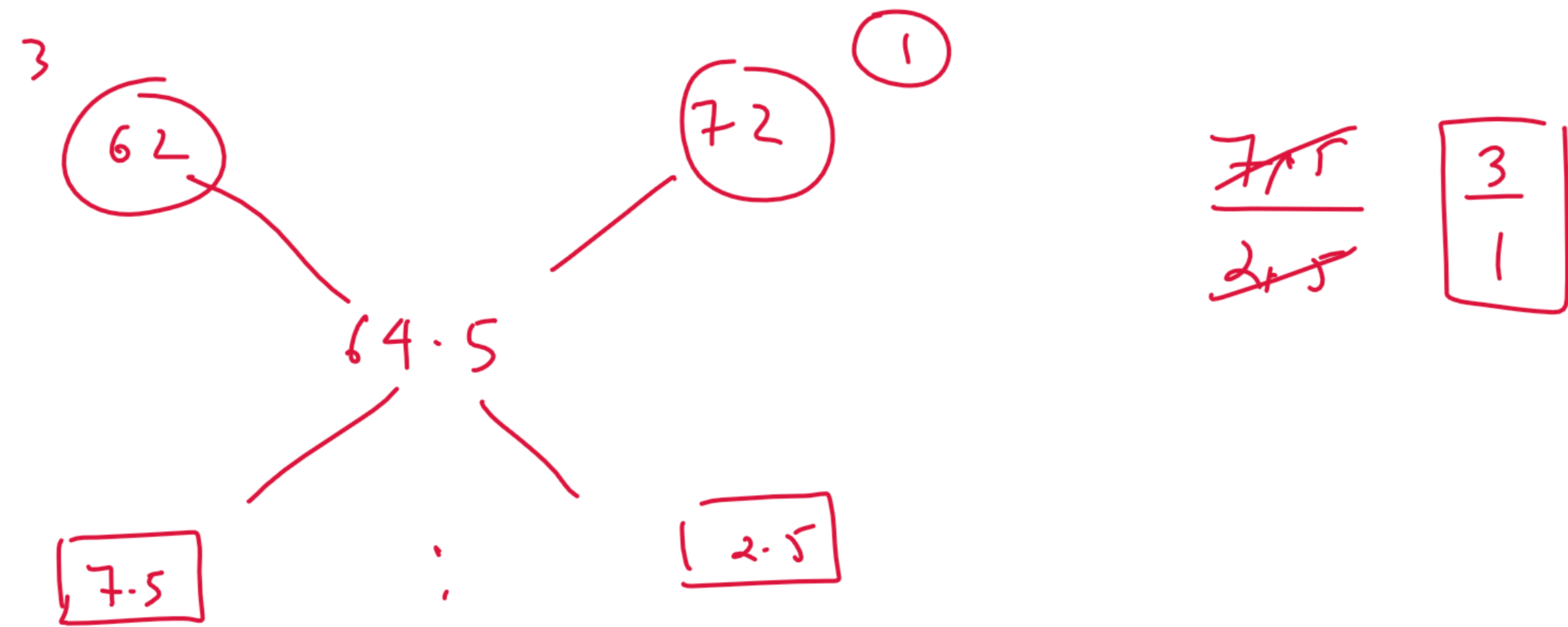
$\star\star$

|                                                      |
|------------------------------------------------------|
| $\text{Dearer} - \text{Mean} = \text{d. of Cheaper}$ |
| $\text{Mean} - \text{Cheaper} = \text{d. of Dearer}$ |

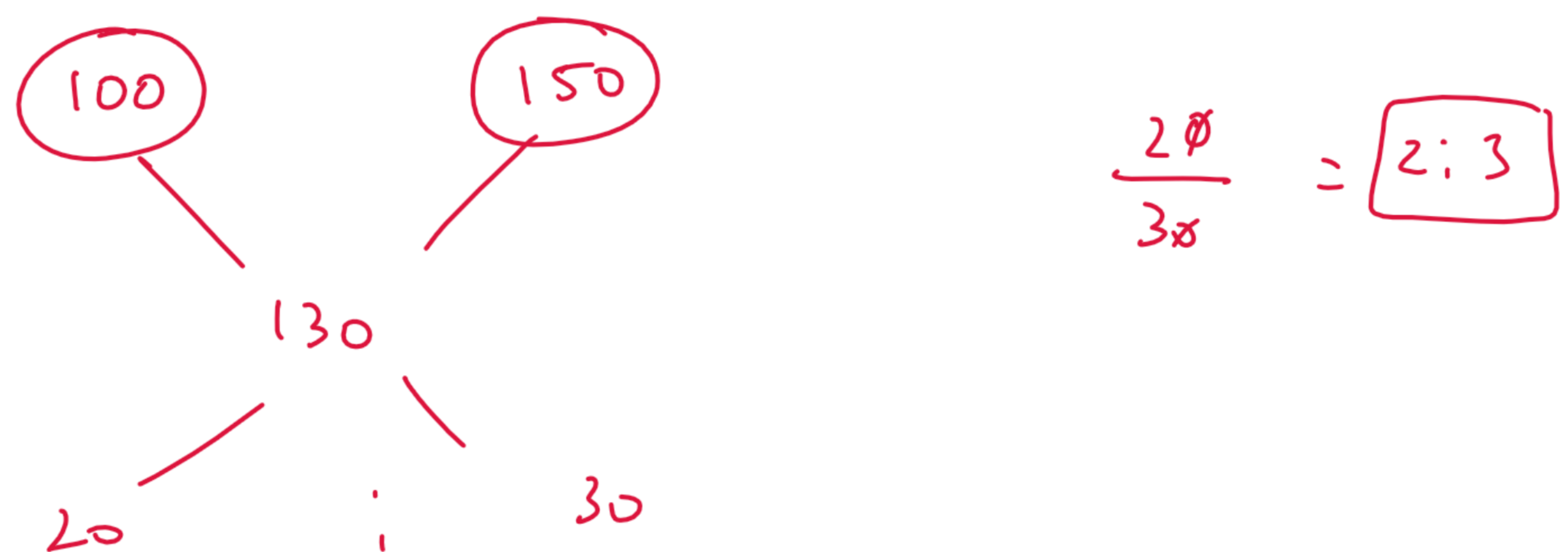


## Mixture and Allegation

- 1) In what ratio must tea Rs. 62 per kg be mixed with tea at Rs. 72 per kg so that the mixture must be worth Rs. 64.50 per kg?



- 2) In what ratio must tea at Rs. 150 per kg be mixed with tea at Rs. 100 per kg so that the mixture must be worth Rs. 130 per kg?





## Mixture and Allegation

In what ratio must wheat at Rs. 3.20 per kg mixed with wheat at Rs. 2.90 per kg so that the mixture be Rs. 3.08 per kg?

In what ratio must water be mixed with milk to gain  $16\frac{2}{3}\%$  on selling the mixture at cost price?

Handwritten solution for the second problem:

Let  $0^w$  and  $100^m$  be the quantities of water and milk respectively.

The cost price of the mixture is  $\frac{600}{7}$ .

The ratio of water to milk is  $1:6$ .

The cost price of 1 litre milk = 100  
 C.P. of 1 litre water = 0  
 C.P. of 1 litre mixture = ?

The selling price of 1 litre mixture = 100

The profit is  $16\frac{2}{3}\%$ .

Calculation:

$$100 = 100 + \frac{50}{3} = \frac{350}{3}$$

Ratio =  $100:1$

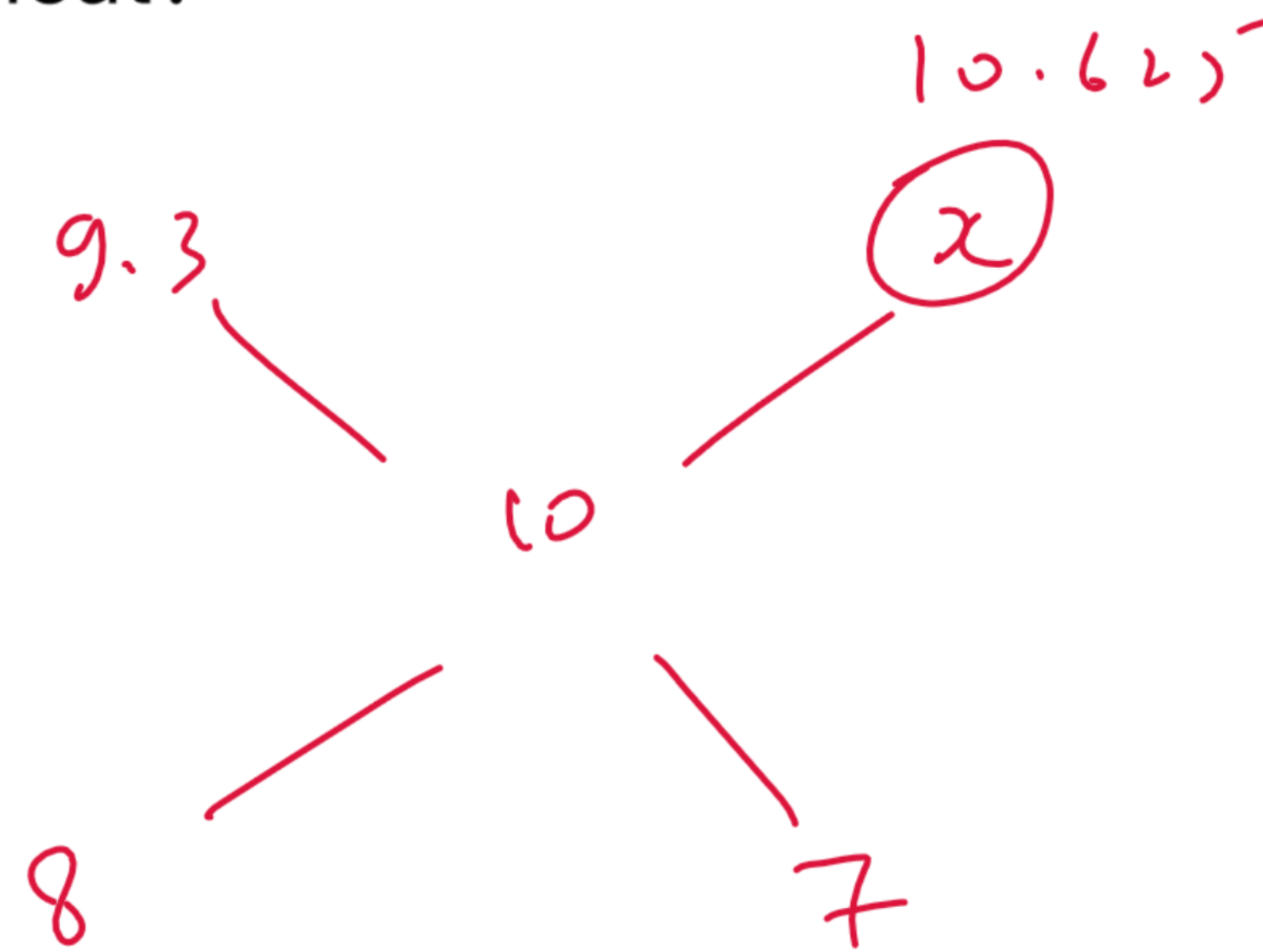
Final calculation:

$$\frac{100 \times 100 \times 2}{3 \times 100} = \frac{600}{7}$$



## Mixture and Allegation

- Q) One quality of wheat at Rs. 9.30 per kg is mixed with another quality at a certain rate in the ratio 8 : 7. If the mixture so formed be worth Rs. 10 per kg, what is the rate per kg of the second quality of wheat?



$$\frac{10 - 9.3}{x - 10} = \frac{8}{7}$$

$$(0.7) \times 7 = 8(x - 10)$$

$$4.9 = 8x - 80$$

$$84.9 = 8x$$

$$\frac{84.9}{8} = x$$

$$\Rightarrow \underline{10.625}$$

A dishonest milkman professes to sell his milk at cost price but he mixes it with water and thereby gains 25%. The percentage of water in the mixture is:

$$\text{C.P of 1 litre} = 100$$

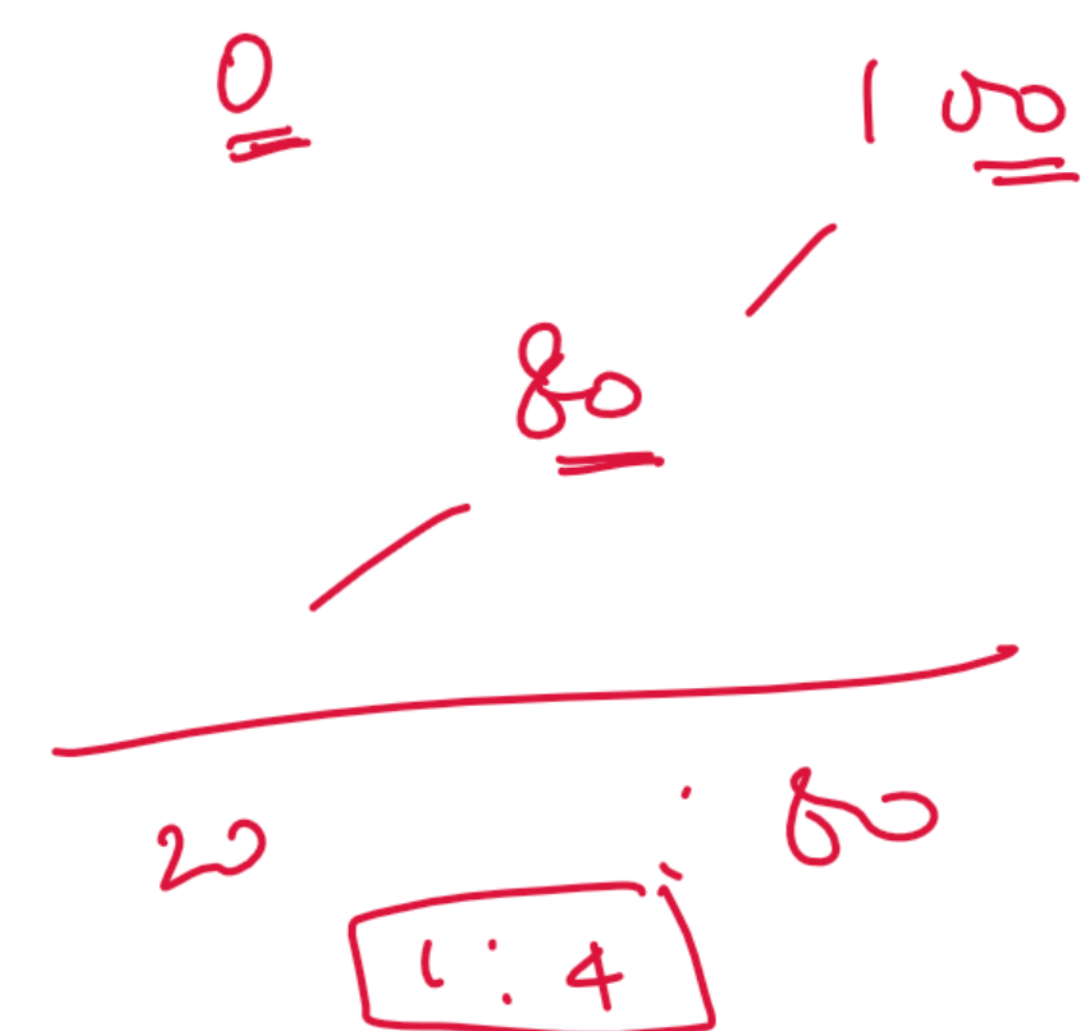
$$\text{C.P of 1 litre} = 0$$

$$\text{S.P of 1 litre} = 100$$

$$100 \Rightarrow 125\%$$

$$x \Rightarrow 100\%$$

$$4 \frac{100 \times 100}{125} = x \Rightarrow \underline{80}$$





## Mixture and Allegation

The price of petrol is Rs. 60 per litre and the price of spirit is Rs. 40 per litre. In what ratio the petrol and spirit be mixed such that the profit after selling the mixture at Rs. 75 per litre be 25%?

$$\text{Petrol} = \underline{\underline{60/\text{litre}}} \quad \text{Spirit} = \underline{\underline{40/\text{litre.}}}$$

$$75 \text{ (q.)} \Rightarrow 125\%$$

$$x \Rightarrow 100\%$$

$$\frac{3 \times 75 + 1 \times 40}{3 + 1} = x$$

$$60 = x$$

$$\begin{array}{ccc} 40 & & 60 \\ & \diagdown & \diagup \\ & 60 & \\ & \diagup & \diagdown \\ \boxed{0} & : & 20 \\ \frac{0}{20} & = & \boxed{0} \end{array}$$

- Q) A sum of Rs. 5000 is lent out in two parts. One at 8% simple interest and the other at 10% simple interest. If the annual interest is Rs. 440, find the sum lent at 8%.

$$\Rightarrow \begin{array}{ccc} \text{8\%} & & 10\% \\ \text{3000} & \text{---} & 2000 \end{array}$$

$$\begin{array}{ccc} & 8.8\% & \\ & \text{---} & \\ 1.2 & & .8 \\ \text{3} & & \text{2} \end{array}$$

$$\frac{3}{5} \times \frac{1000}{1000}$$

$$440 = x\% \text{ of } 5000$$

$$440 = \frac{x}{100} \times 5000$$

$$8.8$$

$$\frac{1120}{180}$$

$$\frac{112}{18} = \frac{3}{2}$$



Mixture and Allegation

8) A sum of Rs. 160 was divided among 40 boys and girls such that each boy received Rs. 5.50 and each girl Rs. 3.50. Find the number of boys and girls.

⇒

Diagram illustrating the mixture and allegation process for the first problem. A central circle contains the number 4. Four boxes are connected to this central circle, containing the values 3.5, 5.5, 1.5, and 0.5. Arrows point from these boxes to two final circles: one containing 30 and another containing 10.

Handwritten calculations for the first problem:

$$\frac{160}{40} = 4$$
$$\frac{3}{3+1} \times 40 = 30$$
$$\frac{1}{3} \times 40 = 13\frac{1}{3}$$
$$\frac{3}{4} \times 40 = 30$$

A sum of Rs. 300 was divided among 100 boys and girls in such a way that each boy gets Rs. 3.60 and each girl gets Rs. 2.40. Find the number of girls.

Handwritten diagram for the second problem showing a central circle with '3' connected to four boxes containing '2.4', '3.6', '0.6', and '0.6'. Arrows point from these boxes to circles containing '50' and '50'.

Handwritten calculations for the second problem:

$$\frac{300}{100} = 3$$
$$\frac{3}{3+1} \times 100 = 75$$
$$\frac{1}{3} \times 100 = 25$$



## Mixture and Allegation

A man travelled a distance of 180 km in 18 hours partly on foot at 8 km/h and partly on bicycle at 17 km/h. Find the distance travelled on foot.

A vessel is filled with liquid, which is 3 parts water and 5 parts milk. How much of the liquid should be drawn off and replaced by water to make it half water and half milk?

$\Rightarrow$  3 parts 5 parts milk = 8 litre 3:5

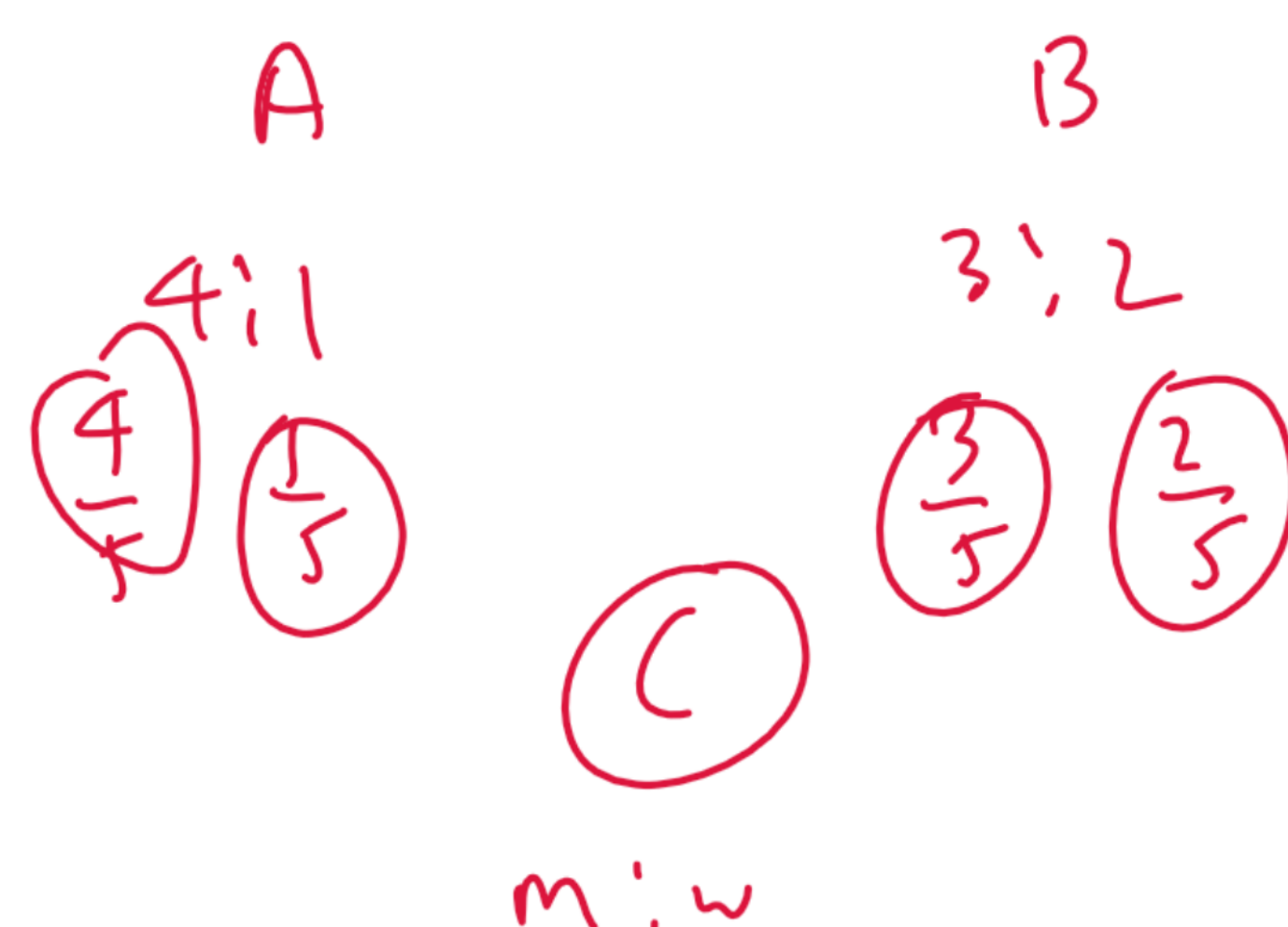
$\boxed{x}$   $\leftarrow \begin{matrix} 3n/8 \\ 5n/8 \end{matrix} \rightarrow$  water

$$3 - \frac{3n}{8} + x = 5 - \frac{5n}{8}$$
$$\frac{24 - 3n + 8n}{8} = \frac{40 - 5n}{8}$$
$$24 + 5n = 40 - 5n$$
$$5n + 5n = 40 - 24$$
$$10n = 16$$
$$\boxed{x = 1.6}$$



## Mixture and Allegation

- Q) Milk and water are in a Can A as 4:1 and in Can B as 3:2. For Can C, if one takes equal quantities from A and B, find the ratio of milk to water in C.



$$\boxed{7:3}$$

$$\frac{4}{5} + \frac{3}{5} : \frac{1}{5} + \frac{2}{5}$$

$$\frac{7}{5} : \frac{3}{5}$$

$$\boxed{7:3}$$

- Q) From the 40 litres solution of pure milk, 5 litres of milk is replaced with equal quality of water. Again 5 litres of the mixture is substituted with 5 litres of water. This operation is repeated one more time. Find the volume of milk in the final solution.

\* Final volume of milk =  $x \left(1 - \frac{a}{x}\right)^n = 40 \left(1 - \frac{5}{40}\right)^3$

$x$  = initial quantity

$a$  = taken out Q.

$n$  = no. of times operation performed

$$40 - 5 = \frac{m}{35} \quad \frac{w}{5}$$

$$7:1$$

$$\frac{m}{7} \quad \frac{w}{8}$$

$$\Rightarrow \boxed{26.7}$$

5 litres  
↑  
m + w

$$64 \overline{) 1715} \\ \underline{-128} \phantom{00} \\ 435 \\ \underline{-384} \phantom{00} \\ 510 \\ \underline{-448} \phantom{00} \\ 62$$

$$\begin{array}{r} 35 \\ \times 49 \phantom{00} 4 \\ \hline 315 \\ 140 \times \phantom{00} 2 \\ \hline 1715 \end{array}$$

$$\frac{40 \times 35 \times 35 \times 35}{40 \times 40 \times 40} = \frac{7}{8} \times \frac{7}{8} \times \frac{7}{8}$$

$$\boxed{40 \text{ lit.}}$$

$$\frac{35 \times 40}{64}$$



## Mixture and Allegation

In two alloys A and B, the ratio of zinc to tin is 5 : 2 and 3 : 4 respectively. How much kg of alloy A should be mixed with 21 kg of the alloy B such that in the final mixture, the quantities of zinc and tin are equal.